3.1 Measure and integration: study notes

These notes condense the module. They follow Folland's order, with Axler's and Tao's viewpoints where those are clearer. Read the texts first; use these notes to consolidate and review. Throughout, m denotes Lebesgue measure and "a.e." means "except on a set of measure zero".

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1. σ-algebras and measures

Definition. A *σ-algebra* on a set X is a collection $\mathcal{M}$ of subsets of X that contains $\emptyset$, is closed under complements, and is closed under countable unions (hence countable intersections). $(X, \mathcal{M})$ is a *measurable space*. For any collection $\mathcal{E}$ of subsets there is a smallest σ-algebra $\sigma(\mathcal{E})$ containing it (the intersection of all σ-algebras containing $\mathcal{E}$). The *Borel σ-algebra* $\mathcal{B}_X$ of a topological space is generated by the open sets. On $\mathbb{R}$, $\mathcal{B}_{\mathbb{R}}$ is also generated by intervals (a, b), or by rays (a, ∞), or by $(-\infty, a]$ with $a \in \mathbb{Q}$.

Definition. A *measure* on $(X, \mathcal{M})$ is a map $\mu : \mathcal{M} \to [0, \infty]$ with $\mu(\emptyset) = 0$ and countable additivity: if $E_1, E_2, \ldots$ are pairwise disjoint, then $\mu\left(\bigcup E_j\right) = \sum \mu(E_j)$.

Basic properties. Monotonicity; countable subadditivity $\mu(\bigcup E_j) \le \sum \mu(E_j)$; *continuity from below*: $E_1 \subseteq E_2 \subseteq \cdots$ implies $\mu(\bigcup E_j) = \lim \mu(E_j)$; *continuity from above*: $E_1 \supseteq E_2 \supseteq \cdots$ and $\mu(E_1) < \infty$ imply $\mu(\bigcap E_j) = \lim \mu(E_j)$.

Proof of continuity from below. Put $F_1 = E_1$, $F_j = E_j \setminus E_{j-1}$. These are disjoint with $\bigcup_{j \le n} F_j = E_n$, so $\mu(\bigcup E_j) = \sum_j \mu(F_j) = \lim_n \mu(E_n)$. For continuity from above, apply this to $E_1 \setminus E_j$ and subtract, which needs $\mu(E_1) < \infty$. (Counterexample without it: $E_n = [n, \infty)$, $m(E_n) = \infty$, $\bigcap E_n = \emptyset$.)

Borel–Cantelli lemma. If $\sum \mu(E_n) < \infty$ then $\mu(\limsup E_n) = 0$, where $\limsup E_n = \bigcap_N \bigcup_{n \ge N} E_n$ is the set of points lying in infinitely many E_n. Proof: $\mu(\limsup E_n) \le \mu(\bigcup_{n \ge N} E_n) \le \sum_{n \ge N} \mu(E_n) \to 0$.

Completeness. μ is *complete* if every subset of a null set is measurable. Every measure has a completion $\overline{\mu}$ on $\overline{\mathcal{M}} = \{E \cup F : E \in \mathcal{M},\ F \subseteq N \in \mathcal{M},\ \mu(N) = 0\}$. Lebesgue measure is the completion of its restriction to Borel sets.

σ-finiteness. μ is σ-finite if X is a countable union of sets of finite measure. Many theorems (Fubini, Radon–Nikodym) need this.

Examples. Counting measure on any set; Dirac measure δ_x; Lebesgue measure; on an uncountable set X, the σ-algebra of countable/co-countable sets with $\mu(E) = 0$ if E is countable and 1 if co-countable.

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2. Outer measures and Carathéodory's theorem

Definition. An *outer measure* on X is $\mu^* : \mathcal{P}(X) \to [0, \infty]$ with $\mu^*(\emptyset) = 0$, monotone, and countably subadditive.

The standard way to get one: given a family $\mathcal{E}$ of "elementary" sets containing $\emptyset$ and X and a function $\rho : \mathcal{E} \to [0, \infty]$ with $\rho(\emptyset) = 0$, set

$$\mu^*(A) = \inf\left\{ \sum_{j=1}^{\infty} \rho(E_j) : E_j \in \mathcal{E},\ A \subseteq \bigcup_j E_j \right\}.$$

For Lebesgue outer measure on $\mathbb{R}^n$, $\mathcal{E}$ is the set of boxes and ρ is volume.

Definition (Carathéodory). $A \subseteq X$ is *μ^*-measurable* if for every $E \subseteq X$,

$$\mu^*(E) = \mu^*(E \cap A) + \mu^*(E \cap A^c).$$

("A splits every set additively.") By subadditivity only $\geq$ needs checking, and only for $\mu^*(E) < \infty$.

Theorem (Carathéodory). The μ^*-measurable sets form a σ-algebra $\mathcal{M}$, and $\mu^*|_{\mathcal{M}}$ is a complete measure.

Proof sketch. Closure under complements is built in. For finite unions: if $A, B \in \mathcal{M}$ then for any E,

$$\mu^*(E) = \mu^*(E\cap A) + \mu^*(E\cap A^c\cap B) + \mu^*(E\cap A^c\cap B^c) \geq \mu^*(E\cap(A\cup B)) + \mu^*(E\cap(A\cup B)^c),$$

using subadditivity on $E \cap (A \cup B) = (E \cap A) \cup (E \cap A^c \cap B)$. So $\mathcal{M}$ is an algebra. Now let $A_j \in \mathcal{M}$ be disjoint, $B_n = \bigcup_{j\leq n} A_j$, $B = \bigcup_j A_j$. Splitting $E \cap B_n$ by A_n gives inductively $\mu^*(E \cap B_n) = \sum_{j\leq n} \mu^*(E \cap A_j)$. Hence

$$\mu^*(E) = \mu^*(E \cap B_n) + \mu^*(E \cap B_n^c) \geq \sum_{j\leq n} \mu^*(E \cap A_j) + \mu^*(E \cap B^c).$$

Let $n \to \infty$ and use subadditivity: $\mu^*(E) \geq \sum_j \mu^*(E \cap A_j) + \mu^*(E \cap B^c) \geq \mu^*(E \cap B) + \mu^*(E \cap B^c)$. So $B \in \mathcal{M}$, and all inequalities are equalities; taking $E = B$ gives countable additivity. Completeness: if $\mu^*(N) = 0$ then $\mu^*(E \cap N) + \mu^*(E \cap N^c) \leq 0 + \mu^*(E)$. □

Extension theorem (Carathéodory–Hahn). If μ_0 is a *premeasure* on an algebra $\mathcal{A}$ (finitely additive, and countably additive whenever a countable disjoint union in $\mathcal{A}$ happens to lie in $\mathcal{A}$), the induced outer measure μ^* restricts to a measure on $\sigma(\mathcal{A})$ extending μ_0. If μ_0 is σ-finite the extension is unique.

The hard step for Lebesgue measure is showing that box volume is a premeasure, i.e. that a box cannot be covered by countably many boxes of smaller total volume. This uses compactness (Heine–Borel): shrink the box slightly to a compact one and enlarge the covering boxes slightly to open ones; finitely many suffice, and the finite case is combinatorial.

Lebesgue–Stieltjes measures. For $F : \mathbb{R} \to \mathbb{R}$ increasing and right-continuous, $\mu_F((a, b]) = F(b) - F(a)$ extends to a unique Borel measure. Every Borel measure on $\mathbb{R}$ finite on bounded sets arises this way. $F(x) = x$ gives Lebesgue measure; a step function gives a sum of point masses; the Cantor function gives a continuous singular measure.

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3. Lebesgue measure: regularity, invariance, non-measurable sets

Regularity. For Lebesgue measurable $E \subseteq \mathbb{R}^n$:

$$m(E) = \inf\{m(U) : U \supseteq E \text{ open}\} = \sup\{m(K) : K \subseteq E \text{ compact}\}.$$

Equivalently, E is measurable iff for every $\varepsilon > 0$ there is open $U \supseteq E$ with $m^*(U \setminus E) < \varepsilon$; iff $E = G \setminus N$ with G a G_δ set and $m(N) = 0$; iff $E = F \cup N$ with F an F_σ set and $m(N) = 0$.

Invariance. $m(E + x) = m(E)$ and $m(TE) = |\det T|\, m(E)$ for linear T. Up to a constant, Lebesgue measure is the only translation-invariant Borel measure on $\mathbb{R}^n$ that is finite on compact sets (a first case of uniqueness of Haar measure).

The Cantor set. $C = \bigcap_k C_k$ where C_k is the union of 2^k closed intervals of length 3^{-k}. Then $m(C) \leq (2/3)^k \to 0$, but C is uncountable: it consists of the points of $[0, 1]$ with a ternary

expansion using only digits 0 and 2, and that set is in bijection with $\{0,1\}^{\mathbb{N}}$. Variations ("fat Cantor sets") remove smaller middle intervals and give closed nowhere dense sets of positive measure.

The Cantor function $F : [0,1] \to [0,1]$ is continuous, increasing, constant on each removed interval (so $F' = 0$ a.e.), and $F(0) = 0$, $F(1) = 1$. So $\int_0^1 F'\,dx = 0 \neq F(1) - F(0)$: the fundamental theorem of calculus fails for merely continuous increasing functions.

Vitali's non-measurable set. Define $x \sim y$ on $[0,1]$ iff $x - y \in \mathbb{Q}$. Using the axiom of choice, pick one point from each class to get V. For the rationals $q_1, q_2, \ldots$ in $[-1,1]$ the translates $V + q_k$ are pairwise disjoint and

$$[0,1] \subseteq \bigcup_k (V + q_k) \subseteq [-1,2].$$

If V were measurable, translation invariance and countable additivity would give $1 \leq \sum_k m(V) \leq 3$, impossible whether $m(V) = 0$ or $m(V) > 0$. So there is no translation-invariant countably additive measure defined on all subsets of $\mathbb{R}$ that gives intervals their length. (Solovay showed that without choice it is consistent that every subset of $\mathbb{R}$ is Lebesgue measurable.)

Also useful: every set of positive outer measure contains a non-measurable subset; and the Lebesgue σ-algebra is strictly larger than the Borel σ-algebra (count: there are $2^{\mathfrak{c}}$ subsets of the Cantor set, all Lebesgue measurable, but only $\mathfrak{c}$ Borel sets).

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4. Measurable functions

Definition. $f : (X, \mathcal{M}) \to (Y, \mathcal{N})$ is measurable if $f^{-1}(B) \in \mathcal{M}$ for every $B \in \mathcal{N}$. It suffices to check this for B in a generating family. For $f : X \to [-\infty, \infty]$ it suffices that $\{f > a\} \in \mathcal{M}$ for every $a \in \mathbb{R}$.

Closure properties. Sums, products, compositions with continuous functions, $\sup_n f_n$, $\inf_n f_n$, $\limsup f_n$, $\liminf f_n$, and pointwise limits of measurable functions are measurable.

Pitfall. "Lebesgue measurable" functions $\mathbb{R} \to \mathbb{R}$ means: preimages of Borel sets are Lebesgue measurable. Then $f \circ g$ with both Lebesgue measurable need **not** be measurable. Compositions are safe if the outer function is Borel.

Simple approximation. If $f : X \to [0, \infty]$ is measurable, there are simple functions $0 \leq \varphi_1 \leq \varphi_2 \leq \cdots \leq f$ with $\varphi_n \to f$ pointwise, and uniformly on any set where f is bounded. Explicitly,

$$\varphi_n = \sum_{k=0}^{n2^n - 1} \frac{k}{2^n} \chi_{\{k2^{-n} \leq f < (k+1)2^{-n}\}} + n\,\chi_{\{f \geq n\}}.$$

Littlewood's three principles (Lebesgue measure, finite-measure sets): every measurable set is nearly a finite union of intervals; every measurable function is nearly continuous (**Lusin**: for $\varepsilon > 0$ there is a closed F with $m(E \setminus F) < \varepsilon$ and $f|_F$ continuous); every convergent sequence is nearly uniformly convergent (**Egorov**).

Egorov's theorem. If $\mu(X) < \infty$ and $f_n \to f$ a.e. (all finite-valued and measurable), then for every $\varepsilon > 0$ there is E with $\mu(E) < \varepsilon$ and $f_n \to f$ uniformly on $X \setminus E$.

Proof. Let $E_n(k) = \bigcup_{j \geq n} \{|f_j - f| \geq 1/k\}$. For fixed k these decrease in n to a null set, so by continuity from above (finite measure!) pick n_k with $\mu(E_{n_k}(k)) < \varepsilon 2^{-k}$. Put $E = \bigcup_k E_{n_k}(k)$. Off E, $|f_j - f| < 1/k$ for all $j \geq n_k$. □

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5. The integral and the convergence theorems

Construction. For simple $\varphi = \sum a_j \chi_{E_j} \geq 0$ (canonical form), $\int \varphi \, d\mu = \sum a_j \mu(E_j)$ with $0 \cdot \infty = 0$. For measurable $f \geq 0$,

$$\int f \, d\mu = \sup \left\{ \int \varphi \, d\mu : 0 \leq \varphi \leq f, \ \varphi \text{ simple} \right\}.$$

(Axler instead defines it as a supremum over finite measurable partitions of $\sum_j \mu(A_j) \inf_{A_j} f$, which is equivalent.) For real f, write $f = f^+ - f^-$; f is *integrable* ($f \in L^1$) if $\int |f| < \infty$, and then $\int f = \int f^+ - \int f^-$. Complex f: integrate real and imaginary parts.

Monotone convergence theorem (MCT). If $0 \leq f_1 \leq f_2 \leq \cdots$ are measurable and $f_n \to f$, then $\int f_n \to \int f$.

Proof. $\int f_n$ increases to some $L \leq \int f$. For the reverse, fix simple $0 \leq \varphi \leq f$ and $0 < c < 1$. The sets $E_n = \{f_n \geq c\varphi\}$ increase to X (if $\varphi(x) > 0$ then $f_n(x) \to f(x) > c\varphi(x)$; if $\varphi(x) = 0$ then $x \in E_n$ for all n). So $\int f_n \geq c \int_{E_n} \varphi \to c \int \varphi$, using continuity from below of the measure $E \mapsto \int_E \varphi$. Let $c \to 1$ and take the sup over φ. $\square$

Corollaries. Integrals of non-negative functions are additive; $\int \sum f_n = \sum \int f_n$ for $f_n \geq 0$; $E \mapsto \int_E f \, d\mu$ is a measure for $f \geq 0$.

Fatou's lemma. If $f_n \geq 0$ are measurable, $\int \liminf f_n \leq \liminf \int f_n$.

Proof. $g_k = \inf_{n \geq k} f_n$ increases to $\liminf f_n$ and $\int g_k \leq \int f_n$ for $n \geq k$; apply MCT. $\square$

Strict inequality can occur: $f_n = \chi_{[n,n+1]}$ ("escape to horizontal infinity"), $f_n = n\chi_{(0,1/n)}$ ("escape to vertical infinity"), $f_n = \frac{1}{n}\chi_{[0,n]}$ ("escape to width infinity"). In each, $f_n \to 0$ pointwise but $\int f_n = 1$. These three examples are worth memorising: every failure of an interchange of limit and integral looks like one of them.

Dominated convergence theorem (DCT). If $f_n \to f$ a.e. and $|f_n| \leq g$ with $g \in L^1$, then $f \in L^1$ and $\int |f_n - f| \to 0$; in particular $\int f_n \to \int f$.

Proof. $2g - |f_n - f| \geq 0$; Fatou gives $\int 2g \leq \liminf \int (2g - |f_n - f|) = \int 2g - \limsup \int |f_n - f|$. Since $\int 2g < \infty$, $\limsup \int |f_n - f| \leq 0$. $\square$

Standard consequences.

- *Differentiation under the integral:* if $f(\cdot, t) \in L^1$, $\partial_t f$ exists and $|\partial_t f(x,t)| \leq g(x) \in L^1$ for all t near t_0, then $\frac{d}{dt} \int f(x,t)\, dx = \int \partial_t f(x,t)\, dx$ (mean value theorem plus DCT).
- *Riemann vs Lebesgue:* a bounded f on $[a,b]$ is Riemann integrable iff it is continuous a.e. (Lebesgue's criterion), and then the two integrals agree. Improper Riemann integrals such as $\int_0^\infty \frac{\sin x}{x}\, dx$ can converge without f being Lebesgue integrable.
- L^1 *as a space:* identify functions equal a.e. Then $\|f\|_1 = \int |f|$ is a norm.

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6. Modes of convergence

On a measure space, $f_n \to f$:

- **a.e.** if $f_n(x) \to f(x)$ off a null set;
- **in measure** if $\mu(\{|f_n - f| \geq \varepsilon\}) \to 0$ for every $\varepsilon > 0$;
- **in** L^p if $\|f_n - f\|_p \to 0$;

- **almost uniformly** if for every ε there is E, $\mu(E) < \varepsilon$, with uniform convergence off E.

Implications.

- L^p convergence $\Rightarrow$ convergence in measure (Chebyshev: $\mu(|f_n - f| \geq \varepsilon) \leq \varepsilon^{-p}\|f_n - f\|_p^p$).
- Almost uniform $\Rightarrow$ a.e. and in measure.
- On finite measure spaces, a.e. $\Rightarrow$ almost uniform (Egorov) $\Rightarrow$ in measure.
- **Riesz:** convergence in measure $\Rightarrow$ some subsequence converges a.e. (choose n_k with $\mu(|f_{n_k} - f| \geq 2^{-k}) < 2^{-k}$ and use Borel–Cantelli).
- DCT upgrades a.e. (or in-measure) convergence to L^1 convergence.

Counterexamples.

- The *typewriter sequence* $\chi_{[j2^{-k},(j+1)2^{-k}]}$ (enumerated over k, then $j = 0, \ldots, 2^k - 1$) tends to 0 in $L^1[0,1]$ and in measure but converges at no point.
- The three escape examples above converge a.e. but not in L^1; $\chi_{[n,n+1]}$ does not even converge in measure (infinite measure space).

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7. Product measures: Tonelli and Fubini

Given $(X, \mathcal{M}, \mu)$ and $(Y, \mathcal{N}, \nu)$, the product σ-algebra $\mathcal{M} \otimes \mathcal{N}$ is generated by rectangles $A \times B$. The premeasure $\pi(A \times B) = \mu(A)\nu(B)$ on finite disjoint unions of rectangles extends to a measure $\mu \times \nu$; it is unique when μ, ν are σ-finite.

Monotone class lemma. If an algebra $\mathcal{A}$ is contained in a monotone class $\mathcal{C}$ (closed under increasing unions and decreasing intersections), then $\sigma(\mathcal{A}) \subseteq \mathcal{C}$. (Alternatively, Dynkin's π-λ theorem.) This is how one shows that a property true for rectangles is true for all of $\mathcal{M} \otimes \mathcal{N}$: for example, that $x \mapsto \nu(E_x)$ is measurable and $(\mu \times \nu)(E) = \int \nu(E_x)\, d\mu(x)$, where $E_x = \{y : (x, y) \in E\}$.

Tonelli's theorem. If μ, ν are σ-finite and $f \geq 0$ is $\mathcal{M} \otimes \mathcal{N}$-measurable, then $x \mapsto \int f(x, y)\, d\nu(y)$ and $y \mapsto \int f(x, y)\, d\mu(x)$ are measurable and

$$\int_{X \times Y} f\, d(\mu \times \nu) = \int_X \left(\int_Y f(x, y)\, d\nu(y) \right) d\mu(x) = \int_Y \left(\int_X f(x, y)\, d\mu(x) \right) d\nu(y).$$

Fubini's theorem. Same conclusion if $f \in L^1(\mu \times \nu)$ (with the inner integrals defined a.e. and integrable).

How to use them. To justify swapping the order for a signed f: apply Tonelli to $|f|$ (always allowed for σ-finite spaces), show one iterated integral of $|f|$ is finite, then apply Fubini to f.

Counterexamples.

- *Not integrable:* on $[0,1]^2$, $f(x, y) = \dfrac{x^2 - y^2}{(x^2 + y^2)^2}$ has iterated integrals $\pi/4$ and $-\pi/4$.
- *Not σ-finite:* μ = Lebesgue on $[0,1]$, ν = counting measure on $[0,1]$, $f = \chi_\Delta$ with Δ the diagonal. Then $\int\int f\, d\mu\, d\nu = 0$ but $\int\int f\, d\nu\, d\mu = 1$.

Lebesgue measure on $\mathbb{R}^n$. m^{n+k} is the completion of $m^n \times m^k$. The completion has a subtlety: for Fubini–Tonelli with complete measures, the slice functions are only measurable for a.e. x. Change of variables: for a C^1 diffeomorphism G between open sets, $\int_{G(\Omega)} f = \int_\Omega (f \circ G)|\det DG|$.

Polar coordinates: $\int_{\mathbb{R}^n} f = \int_0^\infty \int_{S^{n-1}} f(r\omega) r^{n-1}\, d\sigma(\omega)\, dr$.

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8. L^p spaces

For $1 \le p < \infty$, $\|f\|_p = \left(\int |f|^p\right)^{1/p}$; $\|f\|_\infty = \inf\{M : |f| \le M \text{ a.e.}\}$ (the essential supremum). L^p is the space of equivalence classes (equal a.e.) with finite norm. Exponents p, q are *conjugate* if $1/p + 1/q = 1$ (with 1 and ∞ conjugate).

Young's inequality. For $a, b \ge 0$, $ab \le \frac{a^p}{p} + \frac{b^q}{q}$ (concavity of log).

Hölder's inequality. $\|fg\|_1 \le \|f\|_p \|g\|_q$.

Proof. Trivial if either norm is 0 or ∞. Otherwise normalise to $\|f\|_p = \|g\|_q = 1$ and integrate Young's inequality: $\int |fg| \le \frac{1}{p} + \frac{1}{q} = 1$. Equality iff $|f|^p$ and $|g|^q$ are proportional a.e. □

Minkowski's inequality. $\|f + g\|_p \le \|f\|_p + \|g\|_p$.

Proof ($1 < p < \infty$). $|f+g|^p \le (|f| + |g|)|f+g|^{p-1}$. Apply Hölder with exponents p and $q = p/(p-1)$ to each part: $\int |f+g|^p \le (\|f\|_p + \|g\|_p)\|f+g\|_p^{p-1}$. Divide (after checking $\|f+g\|_p < \infty$ via $|f+g|^p \le 2^{p-1}(|f|^p + |g|^p)$). □

Riesz–Fischer theorem. L^p is complete for $1 \le p \le \infty$. Moreover, every L^p-convergent sequence ($p < \infty$) has an a.e.-convergent subsequence.

Proof ($p < \infty$). A normed space is complete iff every absolutely convergent series converges. Let $\sum \|f_k\|_p = B < \infty$. Put $G_n = \sum_{k \le n} |f_k|$ and $G = \sum_k |f_k|$. Minkowski gives $\|G_n\|_p \le B$, and MCT gives $\|G\|_p \le B$, so $G < \infty$ a.e. Hence $F = \sum f_k$ converges absolutely a.e., $|F| \le G \in L^p$, and $|F - \sum_{k \le n} f_k|^p \le (2G)^p \in L^1$. By DCT, $\sum_{k \le n} f_k \to F$ in L^p. □

Inclusions. If $\mu(X) < \infty$ and $p < q$, then $L^q \subseteq L^p$ with $\|f\|_p \le \mu(X)^{1/p - 1/q} \|f\|_q$ (Hölder with $|f|^p \cdot 1$). On ℓ^p (counting measure) the inclusions reverse: $\ell^p \subseteq \ell^q$ for $p < q$. On $\mathbb{R}$, neither inclusion holds.

Density. Simple functions are dense in L^p for $p < \infty$. On $\mathbb{R}^n$, $C_c(\mathbb{R}^n)$ is dense in L^p for $p < \infty$ (by regularity and Urysohn), and so are step functions (finite combinations of indicators of boxes). Consequently translation is continuous: $\|f(\cdot + h) - f\|_p \to 0$ as $h \to 0$ for $p < \infty$. This fails for $p = \infty$ (take $f = \chi_{[0,1]}$).

Duality (preview of 3.2). For σ-finite μ and $1 \le p < \infty$, the map $g \mapsto (f \mapsto \int fg)$ is an isometric isomorphism $L^q \to (L^p)^*$. The proof uses Radon–Nikodym. For $p = \infty$ it is not onto in general.

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9. Signed measures and the Radon–Nikodym theorem

A *signed measure* ν on $(X, \mathcal{M})$ is countably additive, takes values in $(-\infty, \infty]$ or $[-\infty, \infty)$, and $\nu(\emptyset) = 0$. A set P is *positive* if $\nu(E) \ge 0$ for all measurable $E \subseteq P$.

Hahn decomposition. There is a partition $X = P \cup N$ with P positive and N negative; it is unique up to null sets.

Proof idea. Assume ν omits $+\infty$. Let $s = \sup\{\nu(E) : E \text{ positive}\}$, attained by a union P of a maximising sequence. If $N = P^c$ contained a set of positive measure, a "greedy" lemma (every set of positive finite measure contains a positive subset of positive measure, found by repeatedly removing subsets of the most negative available measure) would let you enlarge P. □

Jordan decomposition. $\nu = \nu^+ - \nu^-$ with $\nu^+(E) = \nu(E \cap P)$, $\nu^-(E) = -\nu(E \cap N)$; these are mutually singular, and the decomposition is unique among such pairs. The *total variation* is $|\nu| = \nu^+ + \nu^-$.

Definitions. $\nu \ll \mu$ (*absolutely continuous*) if $\mu(E) = 0 \Rightarrow \nu(E) = 0$. $\nu \perp \mu$ (*mutually singular*) if $X = A \cup B$ disjoint with $\mu(A) = 0 = \nu(B)$.

For finite ν, $\nu \ll \mu$ is equivalent to the ε-δ condition: for every $\varepsilon > 0$ there is $\delta > 0$ with $\mu(E) < \delta \Rightarrow |\nu(E)| < \varepsilon$. (The proof of "⇒" uses Borel–Cantelli. Finiteness matters: $\nu(E) = \int_E x^{-1}\,dx$ on $(0, 1]$ is absolutely continuous but fails the ε-δ condition.)

Lebesgue–Radon–Nikodym theorem. Let μ, ν be σ-finite (positive) measures. Then $\nu = \nu_a + \nu_s$ uniquely with $\nu_a \ll \mu$ and $\nu_s \perp \mu$, and there is a measurable $f \geq 0$, unique μ-a.e., with $\nu_a(E) = \int_E f\,d\mu$. If $\nu \ll \mu$ then $\nu = f\,d\mu$; $f = \frac{d\nu}{d\mu}$ is the *Radon–Nikodym derivative*.

Von Neumann's proof (finite measures). Put $\rho = \mu + \nu$. The functional $\Lambda(g) = \int g\,d\nu$ satisfies $|\Lambda(g)| \leq \nu(X)^{1/2}\|g\|_{L^2(\rho)}$, so by the Riesz representation theorem for the Hilbert space $L^2(\rho)$ there is $h \in L^2(\rho)$ with $\int g\,d\nu = \int gh\,d\rho$ for all $g \in L^2(\rho)$. Testing with indicators shows $0 \leq h \leq 1$ ρ-a.e. Rearranging, $\int g(1-h)\,d\nu = \int gh\,d\mu$. Let $A = \{h < 1\}$, $B = \{h = 1\}$. Taking $g = \chi_B$ gives $\mu(B) = 0$, so $\nu_s = \nu|_B \perp \mu$. On A, take $g = (1 + h + \cdots + h^n)\chi_E$ and let $n \to \infty$ by MCT: $\nu(E \cap A) = \int_E \frac{h}{1-h}\chi_A\,d\mu$. So $f = \frac{h}{1-h}\chi_A$. The σ-finite case follows by partitioning X. □

Failure without σ-finiteness. Lebesgue measure m on $[0, 1]$ is absolutely continuous with respect to counting measure c, but $m = f\,dc$ would force $f(x) = m(\{x\}) = 0$ for all x.

Chain rule. If $\lambda \ll \nu \ll \mu$ then $\frac{d\lambda}{d\mu} = \frac{d\lambda}{d\nu}\frac{d\nu}{d\mu}$ a.e. Probabilists know $\frac{d\nu}{d\mu}$ as a likelihood ratio and as the foundation of conditional expectation.

Complex measures have $|\nu|(E) = \sup \sum_j |\nu(E_j)|$ over measurable partitions; $|\nu|$ is a finite measure, and $d\nu = h\,d|\nu|$ with $|h| = 1$.

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10. Differentiation on $\mathbb{R}^n$

Question: for $f \in L^1_{\text{loc}}(\mathbb{R}^n)$, does $\frac{1}{m(B(x,r))}\int_{B(x,r)} f \to f(x)$ as $r \to 0$?

Vitali covering lemma (finite version). Any finite family of balls has a disjoint subfamily $B_1, \ldots, B_k$ with $\bigcup(\text{all}) \subseteq \bigcup 3B_j$, where $3B$ is the concentric ball of three times the radius. (Greedy: take the largest ball, discard all balls meeting it, repeat.)

Hardy–Littlewood maximal function. $Hf(x) = \sup_{r>0} \frac{1}{m(B(x,r))}\int_{B(x,r)} |f|$. It is lower semicontinuous and satisfies the *weak (1,1) inequality*:

$$m(\{Hf > \alpha\}) \leq \frac{3^n}{\alpha}\|f\|_1.$$

Proof. Let $K \subseteq \{Hf > \alpha\}$ be compact. Each $x \in K$ has a ball B_x with $\int_{B_x} |f| > \alpha m(B_x)$. Take a finite subcover, then a Vitali disjoint subfamily: $m(K) \leq 3^n \sum m(B_j) \leq \frac{3^n}{\alpha}\sum \int_{B_j} |f| \leq \frac{3^n}{\alpha}\|f\|_1$. Use inner regularity. □

(Hf is never in L^1 unless $f = 0$, but H is bounded on L^p for $1 < p \leq \infty$.)

Lebesgue differentiation theorem. If $f \in L^1_{\text{loc}}(\mathbb{R}^n)$, then for a.e. x,

$$\lim_{r \to 0} \frac{1}{m(B(x,r))}\int_{B(x,r)} |f(y) - f(x)|\,dy = 0.$$

Such x are *Lebesgue points*.

Proof. Localise to $f \in L^1$. For continuous g the limit is 0 everywhere. Given ε, choose $g \in C_c$ with $\|f-g\|_1 < \varepsilon$. With $h = f - g$, the lim sup of the averages is at most $Hh(x) + |h(x)|$. So

$$m\left(\{\limsup > 2\alpha\}\right) \le m(\{Hh > \alpha\}) + m(\{|h| > \alpha\}) \le \frac{3^n \varepsilon}{\alpha} + \frac{\varepsilon}{\alpha}.$$

Let $\varepsilon \to 0$, then take the union over $\alpha = 1/k$. □

Consequences.

- *Density points:* for measurable E, a.e. $x \in E$ satisfies $m(E \cap B(x,r))/m(B(x,r)) \to 1$. A measurable set cannot be "half of every interval".
- *FTC, part 1:* if $f \in L^1[a,b]$ and $F(x) = \int_a^x f$, then $F' = f$ a.e.

Functions of one variable.

- *Lebesgue's theorem:* monotone functions, and hence functions of bounded variation (differences of increasing functions), are differentiable a.e.
- $F : [a,b] \to \mathbb{R}$ is *absolutely continuous* (AC) if for every ε there is δ such that $\sum |F(b_i) - F(a_i)| < \varepsilon$ whenever (a_i, b_i) are disjoint with $\sum (b_i - a_i) < \delta$. Lipschitz ⇒ AC ⇒ BV and uniformly continuous.
- **FTC for Lebesgue integrals:** F is AC on $[a,b]$ iff F' exists a.e., $F' \in L^1$, and $F(x) - F(a) = \int_a^x F'$ for all x.
- In measure language: for increasing right-continuous F, the decomposition of μ_F into absolutely continuous, singular continuous and pure point parts corresponds to $F = F_{ac} + F_{sc} + F_{pp}$; F_{ac} is the integral of F'. The Cantor function is purely singular continuous.

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11. Common pitfalls

1. **Using DCT without a dominating function that is independent of n.** The bound must be a single $g \in L^1$. If you cannot find one, try Fatou, MCT, uniform integrability, or a truncation.
2. **Using continuity from above on infinite-measure sets.**
3. **Swapping integrals without checking Tonelli on $|f|$ first.**
4. **Confusing "$f_n \to f$ a.e." with "$\|f_n - f\|_\infty \to 0$".** The latter is uniform convergence off a single null set.
5. **Forgetting that L^p elements are equivalence classes.** "$f(0)$" is meaningless for $f \in L^p$; "f is continuous" means "has a continuous representative".
6. **Assuming Lebesgue measurable sets are Borel**, or that the composition of Lebesgue measurable functions is measurable.
7. **Treating improper Riemann integrals as Lebesgue integrals.** $\int_0^\infty \frac{\sin x}{x}\,dx$ exists as an improper integral, but $\frac{\sin x}{x} \notin L^1(0,\infty)$.
8. **Applying Radon–Nikodým to non-σ-finite measures**, or forgetting that the derivative is only unique a.e.

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12. Connections forward

- **3.2 Functional analysis:** L^p spaces are the basic Banach spaces, and L^2 is the basic Hilbert space. Duality $(L^p)^* = L^q$, weak convergence in L^p, and compactness of integral operators all rest on this module. The Riesz–Markov theorem identifies $C(K)^*$ with finite regular Borel measures (Folland ch. 7).
- **Probability:** a probability space is a measure space of total mass 1; random variables are measurable functions; expectation is the integral; conditional expectation is a Radon–Nikodým derivative; the Borel–Cantelli lemma and modes of convergence are used con-

stantly. Kolmogorov's extension theorem is a Carathéodory construction on an infinite product.

- **Harmonic analysis and PDE:** the maximal function and Lebesgue differentiation are the start of Calderón–Zygmund theory; L^p and Fubini underlie convolution, Fourier transforms and Sobolev spaces.
- **Geometry:** Hausdorff measure (Stein–Shakarchi ch. 7, Folland §11.2) defines fractal dimension. Haar measure on locally compact groups generalises Lebesgue measure and is needed in representation theory of compact groups (compare 3.3, where averaging over a finite group proves Maschke's theorem).
- **3.5 Smooth manifolds:** integration of differential forms reduces, in charts, to Lebesgue integrals on $\mathbb{R}^n$ and the change-of-variables formula; Sard's theorem is a measure-zero statement.